Habib University
Introduction to Probability and Statistics – EE 354/CE 361/MATH 310 – Spring 2026
Instructor – Dr. Aamir Hasan - Section L4

Time = 20 Minutes **Quiz No 07** [20 points]

Provide all your answers on the answer sheet. Present your final answers and reasoning clearly. Most of the total score for the questions will be awarded on your reasoning.

CLO Assessed – CLO No 04 - Construct probability distributions, up to two dimensions, from given data for discrete and continuous random variables. (Cog - 3)

Question 1 [12 points] [Cumulative Distribution Function for a Mixed Random Variable]

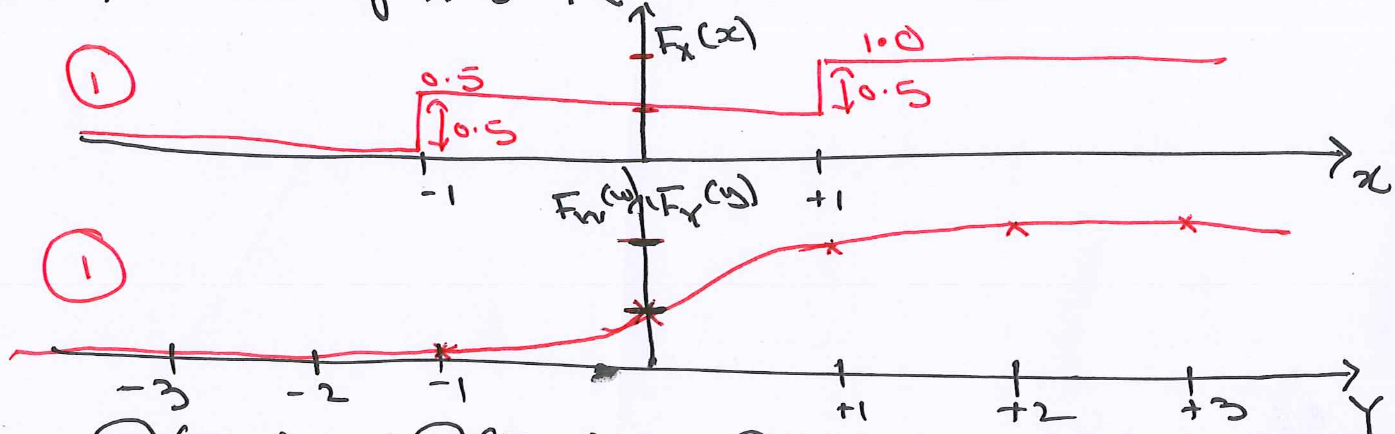
- **Discrete Random Variable X - RV X** is +1 or -1 with probability 0.5.
- **Continuous Random Variable Y –** RV Y is a standard Normal random variable.
- **Continuous Random Variable W –** RV W is also a standard Normal random variable.
- **Random Variable Z = X** with probability 1/3, **Z = Y** with probability 1/3 and **Z = W** with probability 1/3.

*Note – Random Variable **Z** has a mixed distribution.*

Part a [08 points] Calculate and plot the Cumulative Distribution Function (CDF) for RV **Z**. ***Use the diagram given below to plot the CDF, clearly marking the exact CDF heights.***

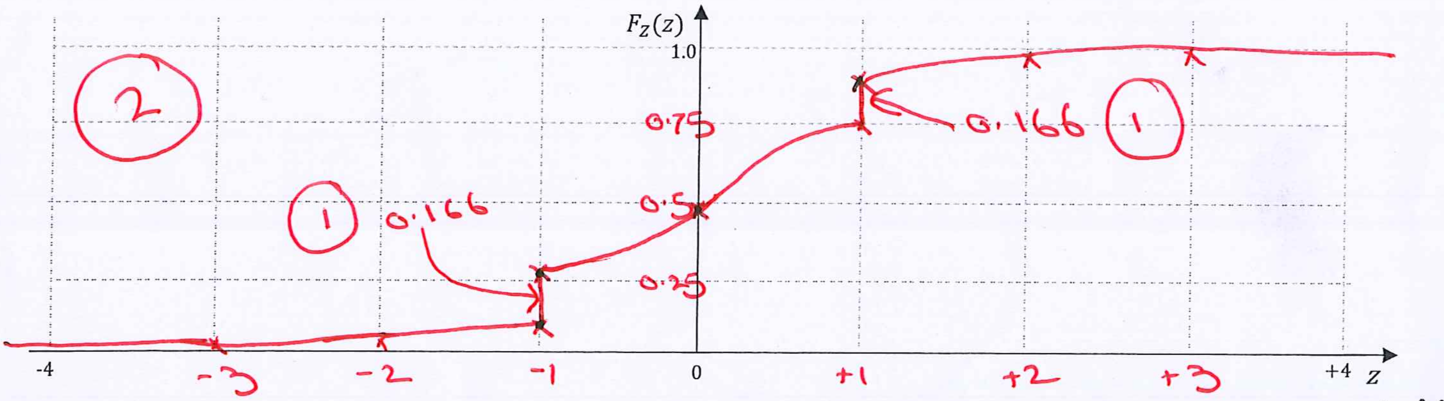
Part b [04 points] Calculate the $P(-1 < Z \le +1)$ using the CDF plot in part a.

a) Lets plot CDF of X & Y (Y is similar to W)



$F_Z(z) = P(Z \le z) = 1/3\, P(X \le z) + 1/3\, P(Y \le z) + 1/3\, P(W \le z)$

$= 1/3\, F_X(z) + 2/3\, F_Y(z) \rightarrow$ also $F_Z(z) = 1/3\, F_X(x) + 2/3\, F_W(z)$ 2



2

$F_X(-1) = 1/2 \Rightarrow 1/3\, F_X(-1) = 1/6 = 0.166$

$F_Y(-1) = 0.1587, \Rightarrow 2/3\, F_Y(-1) = 0.105$

$\Downarrow$

$F_Z(-1) = 0.166 + 0.105 = 0.2718$

$1/3\, F_X(0) = 0.166,\ F_Y(0) = 0.5 \Rightarrow 2/3\, F_Y(0) = 1/3$

$\Rightarrow F_Z(0) = 0.1666 + 0.333 = 0.49$

$1/3\, F_X(+1) = 1/3 \cdot 1 = 0.333$

$F_Y(+1) = 0.8413, \Rightarrow 2/3\, F_Y(+1) = 0.560$

$F_Z(+1) = 0.333 + 0.560 = 0.894$

(b) $P(-1 < Z \leq +1)$

$$F_Z(+1) - F_Z(-1)$$

$$\downarrow$$

1 1

$$0.894 - 0.2718$$

$$= 0.62 \rightarrow 2$$

CLO Assessed – **CLO No 04 - Construct probability distributions, up to two dimensions, from given data for discrete and continuous random variables. (Cog - 3)**

Question 2 [08 points] [Cumulative Distribution Function for a Normal Random Variable]

The Random Variable $Y = X + W$, where $W \sim N(0,1)$ is a Standard Normal random variable (standard normal table is provided). R.V X, independent of W, takes on values +3 or −3 with equal probability of 0.5. Using suitable methods and arguments, determine and plot the CDF of the Random Variable Y, (use the bottom plot given below).

Note – while exact CDF values are not required for the entire range but the ones that are critical (to show your understanding and correctness) must be captured. Include enough details, labelling and arguments by showing critical & needed values on the x-axis and y-axis. ***Hint*** - try to first get a grip on the conditional Y and then determine the CDF of Y.

Given $X = +3$ or -3, $Y|X \sim N(+3,1)$ or $N(-3,1)$

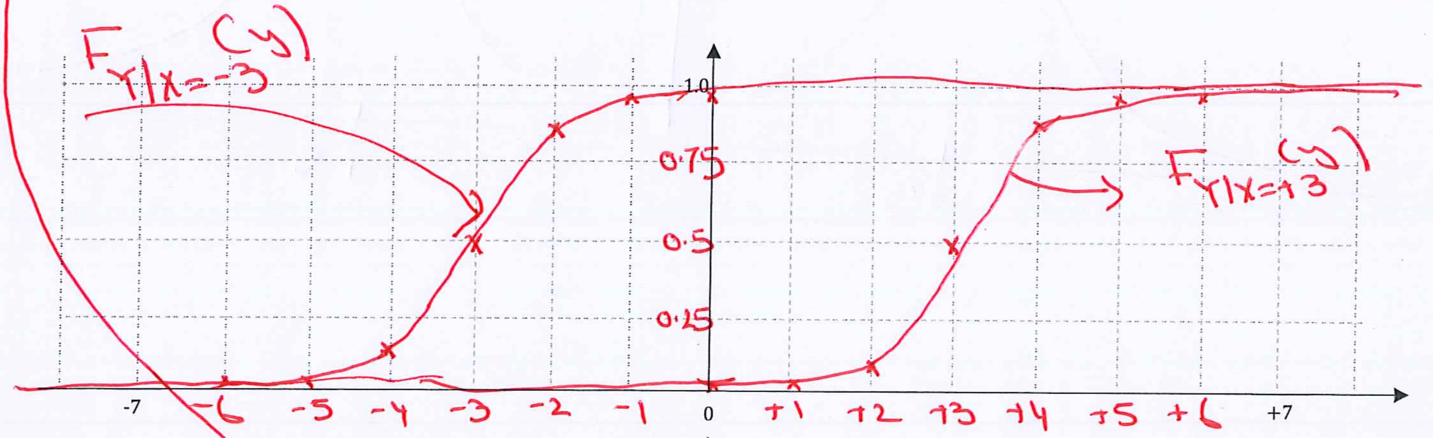
$\Rightarrow Y|X=+3 \sim N(+3,1)$ & $Y|(X=-3) \sim N(-3,1)$ Since $W \sim N(0,1)$

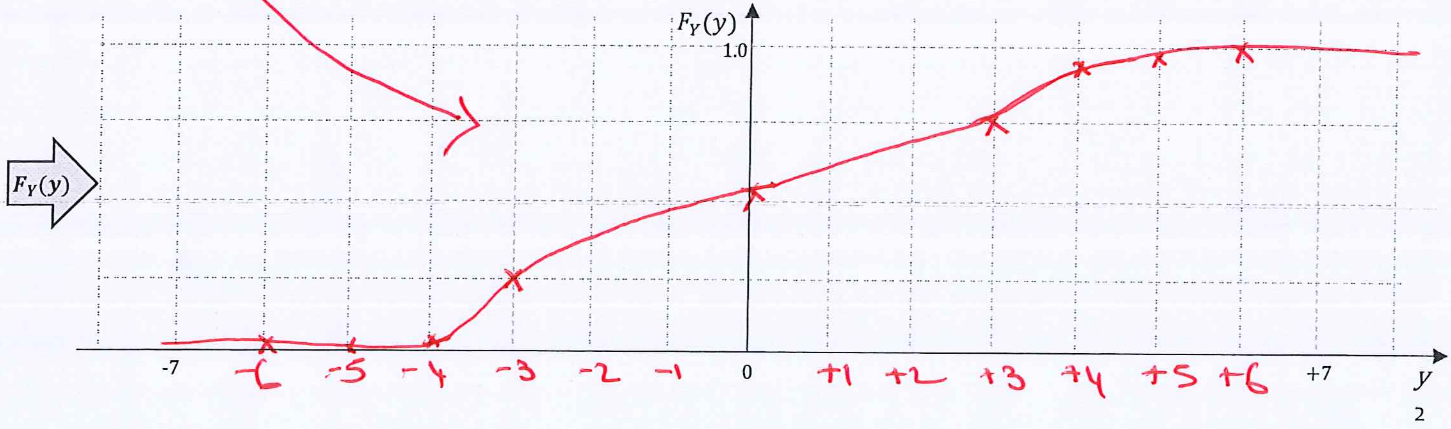
$\Rightarrow f_Y(y) = f_{Y|X=+3}(y) \cdot \frac{1}{2} + f_{Y|X=-3}(y) \cdot \frac{1}{2}$

also, $F_Y(y) = \frac{1}{2} F_{Y|X=+3}(y) + \frac{1}{2} F_{Y|X=-3}(y)$ — (A)

$P(Y \leq y) = \frac{1}{2} P(Y \leq y | X=+3) + \frac{1}{2} P(Y \leq y | X=-3)$

lets plot $F_{Y|X=+3}(y)$ & $F_{Y|X=-3}(y)$





This is not asked ~~and~~ nor required

lets see the plots for conditional pdfs